EXP5: use DATASet in CSV. The containing information About books.

Date :

To perform Exploratory DATA visualization

1. Import libraries.
2. loading Data set.
3. Plot bar graph, scatter, plot box, plot, Histogram., line graph.

Aim:- To use dataset in CSV file containing information about books to perform exploratory Data visualization

Import format.

Book ID	Title	Author	pages	price	Rating	year
1	python Basic	John	250	350	4.5	2020
2	ML Guide	Smith	400	550	4.8	2021
3	DataScience	Alex	320	450	4.3	2019
4	Neural Network	Smith	415	460	4.5	2020
5	Deeplearning	Morland	350	390	4.6	2022
6	NLP	Michale	225	480	3.9	2023

1. Import libraries.

```
import pandas as pd
import numpy as np
import matplotlib pyplot as plt.
```

2. loading Dataset:-

```
data = {
    "Book id": [1, 2, 3, 4, 5, 6],
    "title": ["python, ML Guide, Data science, DL, NLP, NN"],
    "Author": ["John", "Smith", "Alex", "smith", "Mayland", "michale"],
    "pages": [250, 400, 320, 415, 350, 235],
    "price": [350, 550, 450, 460, 390, 480],
    "Rating" [4.5, 4.8, 4.3, 4.5, 4.6, 3.9],
    "Year": ["2020, 2021, 2019, 2020, 2021, 2023]

df = pd.Data frame (data).
df.to_csv ("books.csv, index = false).
print (df).
```

3. plot bar graph, Scatter plot, Boxplot, Histogram & line graph.

```
#1. Bar Graph
plt.figure (fig size = (10, 6))
plt.bar (df ['title'], df ['price']).
plt.title ('book title vs price')
plt.y label ('title')
plt.x label ("price").
plt.show()
```

2. Scatter plot.

```
plt.figure(figsize=(10,6))
plt.scatter(df['pages'], df['price'])
plt.title('pages vs price')

plt.xlabel('pages')
plt.ylabel('price')

plt.show()
```

3. Box plot:-

```
plt.figure(figsize=(10,6))
plt.boxplot(df['price'])

plt.title('Box plot of Book price')
plt.ylabel('price')
plt.show()
```

#4 Histograms:-

```
plt.figure (fig size (10,6)).
plt.hist (df ['Rating']).

plt.title ('Histogram of Rating")

plt.ylabel ("Rating").
plt.xlabel ('frequency').
plt.show()
```

#5 Line graph:

```
plt.figure (fig size = (10, 0))
plt.plot (df ['Year'], df ['Rating"])
plt.title ("Year as Rating

plt.ylabel (" Year)
plt.xlabel ("Rating")

plt.show().
```